

2.2 Methods of accounting for component losses

The performance of real cycles differs from that of ideal cycles for the following reasons:

- (a) Because fluid velocities are high in turbomachinery the change in kinetic energy between the inlet and outlet of each component cannot necessarily be ignored. A further consequence is that the compression and expansion processes are irreversible adiabatics and therefore involve an increase in entropy.
- (b) Fluid friction results in pressure losses in combustion chambers and heat-exchangers, and also in the inlet and exhaust ducts. (Losses in the ducts connecting components are often included in the associated component losses.)
- (c) If a heat-exchanger is to be of economic size, terminal temperature differences are inevitable, i.e. the compressed air cannot be heated to the temperature of the gas leaving the turbine.
- (d) Slightly more work than that required for the compression process will be necessary to overcome bearing and 'windage' friction in the transmission between compressor and turbine, and to drive ancillary components such as fuel and oil pumps.
- (e) The values of c_p and γ of the working fluid vary throughout the cycle owing to changes of temperature and, with internal combustion, to changes in chemical composition.
- (f) The definition of the efficiency of an ideal cycle is unambiguous, but this is not the case for an open cycle with internal combustion. Knowing the compressor delivery temperature, composition of the fuel, and turbine inlet temperature required, a straightforward combustion calculation yields the fuel/air ratio necessary; and a combustion efficiency can also be included to allow for incomplete combustion. Thus it will be possible to express the cycle performance unambiguously in terms of fuel consumption per unit net work output, i.e. in terms of the *specific fuel consumption*. To convert this to a cycle efficiency it is necessary to adopt some convention for expressing the heating value of the fuel.
- (g) With internal combustion, the mass flow through the turbine might be thought to be greater than that through the compressor by virtue of the fuel added. In practice, about 1–2 per cent of the compressed air is bled off for cooling turbine discs and blade roots, and we shall see later that the fuel/air ratio employed is in the region 0.01–0.02. For many cycle calculations it is sufficiently accurate to assume that the fuel added merely compensates for this loss. We will assume in the numerical examples which follow that the mass flows through the compressor and turbine are equal. When turbine inlet temperatures higher than about 1350 K are used, the turbine blading must be internally cooled as well as the disc and blade roots. We then have what is called an *air-cooled* turbine. Up to 15 per cent of the compressor delivery air

might be bled for cooling purposes, and for an accurate estimate of cycle performance it is necessary to account explicitly for the variation of mass flow through the engine.

Methods of accounting for factors (a) to (f) must be discussed before giving examples of cycle calculations. We will also say a little more in general terms about cooling bleed flows.

Stagnation properties

The kinetic energy terms in the steady flow energy equation can be accounted for implicitly by making use of the concept of *stagnation* (or *total*) enthalpy. Physically, the stagnation enthalpy h_0 is the enthalpy which a gas stream of enthalpy h and velocity C would possess when brought to rest adiabatically and without work transfer. The energy equation then reduces to

$$(h_0 - h) + \frac{1}{2}(0 - C^2) = 0$$

and thus h_0 is defined by

$$h_0 = h + C^2/2 \quad (2.6)$$

When the fluid is a perfect gas, $c_p T$ can be substituted for h , and the corresponding concept of *stagnation* (or *total*) *temperature* T_0 is defined by

$$T_0 = T + C^2/2c_p \quad (2.7)$$

$C^2/2c_p$ is called the *dynamic temperature* and, when it is necessary to emphasize the distinction, T is referred to as the *static temperature*. An idea of the order of magnitude of the difference between T_0 and T is obtained by considering air at atmospheric temperature, for which $c_p = 1.005 \text{ kJ/kg K}$, flowing at 100 m/s. Then

$$T_0 - T = \frac{100^2}{2 \times 1.005 \times 10^3} \approx 5 \text{ K}$$

It follows from the energy equation that if there is no heat or work transfer T_0 will remain constant. If the duct is varying in cross-sectional area, or friction is degrading directed kinetic energy into random molecular energy, the static temperature will change—but T_0 will not. Applying the concept to an adiabatic compression, the energy equation becomes

$$W = -c_p(T_2 - T_1) - \frac{1}{2}(C_2^2 - C_1^2) = -c_p(T_{02} - T_{01})$$

Similarly for a heating process without work transfer,

$$Q = c_p(T_{02} - T_{01})$$

Thus if stagnation temperatures are employed there is no need to refer explicitly to the kinetic energy term. A practical advantage is that it is easier to

measure the stagnation temperature of a high-velocity stream than the static temperature (see section 6.5).

When a gas is slowed down and the temperature rises, there is a simultaneous rise in pressure. The *stagnation* (or *total*) *pressure* p_0 is defined in a similar way to T_0 but with the added restriction that the gas is imagined to be brought to rest not only adiabatically but also reversibly, i.e. isentropically. The stagnation pressure is thus defined by

$$\frac{p_0}{p} = \left(\frac{T_0}{T} \right)^{\gamma/(\gamma-1)} \quad (2.8)$$

Stagnation pressure, unlike stagnation temperature, is constant in a stream flowing without heat or work transfer only if friction is absent: the drop in stagnation pressure can be used as a measure of the fluid friction.

It is worth noting that p_0 is not identical with the usual pitot pressure p_0^* defined for incompressible flow by

$$p_0^* = p + \rho C^2/2$$

Substituting equation (2.7) in (2.8), and making use of $c_p = \gamma R/(\gamma - 1)$ and $p = \rho RT$, we have

$$p_0 = p \left(1 + \frac{\rho C^2}{2p} \times \frac{\gamma - 1}{\gamma} \right)^{\gamma/(\gamma-1)}$$

p_0^* is seen to be given by the first two terms of the binomial expansion. Thus p_0 approaches p_0^* as the velocity is decreased and compressibility effects become negligible. As an example of the difference at high velocities, for air moving with sonic velocity (Mach number $M = 1$), $p_0/p = 1.89$ whereas $p_0^*/p = 1.7$. Thus by assuming the flow to be incompressible the stagnation pressure would be underestimated by about 11 per cent.

Applying equation (2.8) to an isentropic compression between inlet 1 and outlet 2 we have the stagnation pressure ratio given by

$$\frac{p_{02}}{p_{01}} = \frac{p_{02}}{p_2} \times \frac{p_1}{p_{01}} \times \frac{p_2}{p_1} = \left(\frac{T_{02}}{T_2} \times \frac{T_1}{T_{01}} \times \frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)} = \left(\frac{T_{02}}{T_{01}} \right)^{\gamma/(\gamma-1)}$$

Similarly, if required,

$$\frac{p_{02}}{p_1} = \left(\frac{T_{02}}{T_1} \right)^{\gamma/(\gamma-1)}$$

Thus p_0 and T_0 can be used in the same way as static values in isentropic $p - T$ relations. Stagnation pressure and temperature are properties of the gas stream which can be used with static values to determine the combined thermodynamic and mechanical state of the stream. Such state points can be represented on the $T-s$ diagram, as shown in Fig. 2.9, which depicts a compression process between 'static' states 1 and 2; the differences between the constant p and p_0 lines have been exaggerated for clarity. The ideal stagnation state which would be reached after isentropic

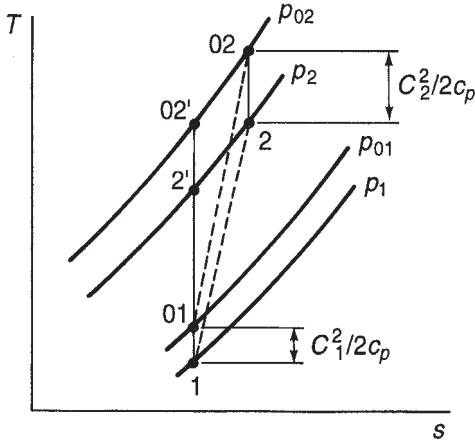


FIG. 2.9 Stagnation states

compression to the same, actual, outlet stagnation pressure is indicated by 02'. Primes will be attached to symbols to denote such ideal states throughout this book.

Compressor and turbine efficiencies

(a) Isentropic efficiency

The efficiency of any machine, the object of which is the absorption or production of work, is normally expressed in terms of the ratio of actual and ideal work transfers. Because turbomachines are essentially adiabatic, the ideal process is isentropic and the efficiency is called an *isentropic efficiency*. Making use of the concept of stagnation enthalpy or temperature to take account of any change in kinetic energy of the fluid between inlet and outlet, we have, for the compressor,

$$\eta_c = \frac{W'}{W} = \frac{\Delta h'_0}{\Delta h_0}$$

For a perfect gas $\Delta h_0 = c_p \Delta T_0$ and this relation is usually sufficiently accurate for real gases under conditions encountered in gas turbines if a mean c_p over the relevant range of temperature is used—see under heading ‘Variation of specific heat’ below for further discussion of this point. Furthermore, because the ideal and actual temperature changes are not very different, the mean c_p can be assumed the same for both so that the compressor isentropic efficiency is normally *defined* in terms of temperature as

$$\eta_c = \frac{T'_{02} - T_{01}}{T_{02} - T_{01}} \quad (2.9)$$

Similarly the turbine isentropic efficiency is defined as

$$\eta_t = \frac{W}{W'} = \frac{T_{03} - T_{04}}{T_{03} - T'_{04}} \quad (2.10)$$

When performing cycle calculations, values of η_c and η_t will be assumed and the temperature equivalents of the work transfers for a given pressure ratio are then found as follows:

$$T_{02} - T_{01} = \frac{1}{\eta_c}(T'_{02} - T_{01}) = \frac{T_{01}}{\eta_c} \left(\frac{T'_{02}}{T_{01}} - 1 \right)$$

and finally

$$T_{02} - T_{01} = \frac{T_{01}}{\eta_c} \left[\left(\frac{p_{02}}{p_{01}} \right)^{(\gamma-1)/\gamma} - 1 \right] \quad (2.11)$$

Similarly

$$T_{03} - T_{04} = \eta_t T_{03} \left[1 - \left(\frac{1}{p_{03}/p_{04}} \right)^{(\gamma-1)/\gamma} \right] \quad (2.12)$$

When the compressor is part of a stationary gas turbine, having a short intake fairing that can be regarded as part of the compressor, p_{01} and T_{01} in equation (2.11) will be equal to p_a and T_a respectively because the velocity of the ambient air is zero. This will be assumed to be the case throughout this chapter. Industrial gas turbine plant usually incorporates a long inlet duct and air filter, however, in which case an inlet pressure loss (Δp_i) must be deducted, i.e. p_{01} will be $(p_a - \Delta p_i)$. Inlet losses vary with installation configurations and engine manufacturers often quote performance figures for zero inlet loss, with corrections for different levels of loss. The situation is rather different when the compressor is part of an aircraft propulsion unit because then there will be an intake duct of significant length in which ram compression takes place due to the forward speed of the aircraft. In this situation p_{01} and T_{01} would differ from p_a and T_a even if there were no friction losses, and it is always necessary to consider the intake and compressor as separate components. A discussion of how intake losses are then taken into account will be deferred until the next chapter.

In defining η_t according to (equation (2.10)) and thus taking the ideal work as proportional to $(T_{03} - T'_{04})$, we are implying that the kinetic energy in the exhaust gas is going to be utilized, e.g. in a subsequent turbine or in the propelling nozzle of a jet engine. But if the turbine is part of an industrial plant exhausting directly to atmosphere, this kinetic energy is wasted. The ideal quantity of turbine work would then seem to be taken more appropriately as that produced by an isentropic expansion from p_{03} to the static outlet pressure p_4 , with p_4 equal to the ambient pressure p_a . Thus η_t would be defined by

$$\eta_t = \frac{T_{03} - T_{04}}{T_{03} \left[1 - \left(\frac{1}{p_{03}/p_a} \right)^{(\gamma-1)/\gamma} \right]} \quad (2.13)$$

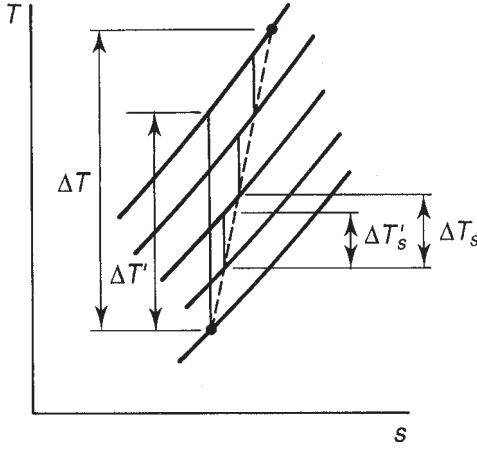


FIG. 2.11

Consider an axial flow compressor consisting of a number of successive stages. If the blade design is similar in successive blade rows it is reasonable to assume that the isentropic efficiency of a single stage, η_s , remains the same through the compressor. Then the overall temperature rise can be expressed by

$$\Delta T = \sum \frac{\Delta T'_s}{\eta_s} = \frac{1}{\eta_s} \sum \Delta T'_s$$

Also, $\Delta T = \Delta T' / \eta_c$ by definition of η_c , and thus

$$\frac{\eta_s}{\eta_c} = \frac{\sum \Delta T'_s}{\Delta T'}$$

But, because the vertical distance between a pair of constant pressure lines in the T - s diagram increases as the entropy increases, it is clear from Fig. 2.11 that $\sum \Delta T'_s > \Delta T'$. It follows that $\eta_c < \eta_s$ and that the difference will increase with the number of stages, i.e. with increase of pressure ratio. A physical explanation is that the increase in temperature due to friction in one stage results in more work being required in the next stage: it might be termed the 'preheat' effect. A similar argument can be used to show that for a turbine $\eta_t > \eta_s$. In this case frictional 'reheating' in one stage is partially recovered as work in the next.

These considerations have led to the concept of *polytropic* (or *small-stage*) efficiency η_{∞} , which is defined as the isentropic efficiency of an elemental stage in the process such that it is constant throughout the whole process. For a compression,

$$\eta_{\infty} = \frac{dT'}{dT} = \text{constant}$$

But $T/p^{(\gamma-1)/\gamma} = \text{constant}$ for an isentropic process, which in differential form is

$$\frac{dT'}{T} = \frac{\gamma - 1}{\gamma} \frac{dp}{p}$$

Substitution of dT' from the previous equation gives

$$\eta_{\infty c} \frac{dT}{T} = \frac{\gamma - 1}{\gamma} \frac{dp}{p}$$

Integrating between inlet 1 and outlet 2, with $\eta_{\infty c}$ constant by definition, we have

$$\eta_{\infty c} = \frac{\ln(p_2/p_1)^{(\gamma-1)/\gamma}}{\ln(T_2/T_1)} \quad (2.14)$$

This equation enables $\eta_{\infty c}$ to be calculated from measured values of p and T at the inlet and outlet of a compressor. Equation (2.14) can also be written in the form

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma \eta_{\infty c}} \quad (2.15)$$

Finally, the relation between $\eta_{\infty c}$ and η_c is given by

$$\eta_c = \frac{T'_2/T_1 - 1}{T_2/T_1 - 1} = \frac{(p_2/p_1)^{(\gamma-1)/\gamma} - 1}{(p_2/p_1)^{(\gamma-1)/\gamma \eta_{\infty c}} - 1} \quad (2.16)$$

Note that if we write $(\gamma - 1)/\gamma \eta_{\infty c}$ as $(n - 1)/n$, equation (2.15) is the familiar relation between p and T for a polytropic process, and thus the definition of η_{∞} implies that the non-isentropic process is polytropic. This is the origin of the term polytropic efficiency.

Similarly, since $\eta_{\infty t}$ is dT/dT' , it can be shown that for an expansion between inlet 3 and outlet 4,

$$\frac{T_3}{T_4} = \left(\frac{p_3}{p_4} \right)^{\eta_{\infty t}(\gamma-1)/\gamma} \quad (2.17)$$

and

$$\eta_t = \frac{1 - \left(\frac{1}{p_3/p_4} \right)^{\eta_{\infty t}(\gamma-1)/\gamma}}{1 - \left(\frac{1}{p_3/p_4} \right)^{(\gamma-1)/\gamma}} \quad (2.18)$$

Making use of equations (2.16) and (2.18) with $\gamma = 1.4$, Fig. 2.12 has been drawn to show how η_c and η_t vary with pressure ratio for a fixed value of polytropic efficiency of 90 per cent in each case.

In practice, as with η_c and η_t , it is normal to define the polytropic efficiencies in terms of stagnation temperatures and pressures. Furthermore, when employing

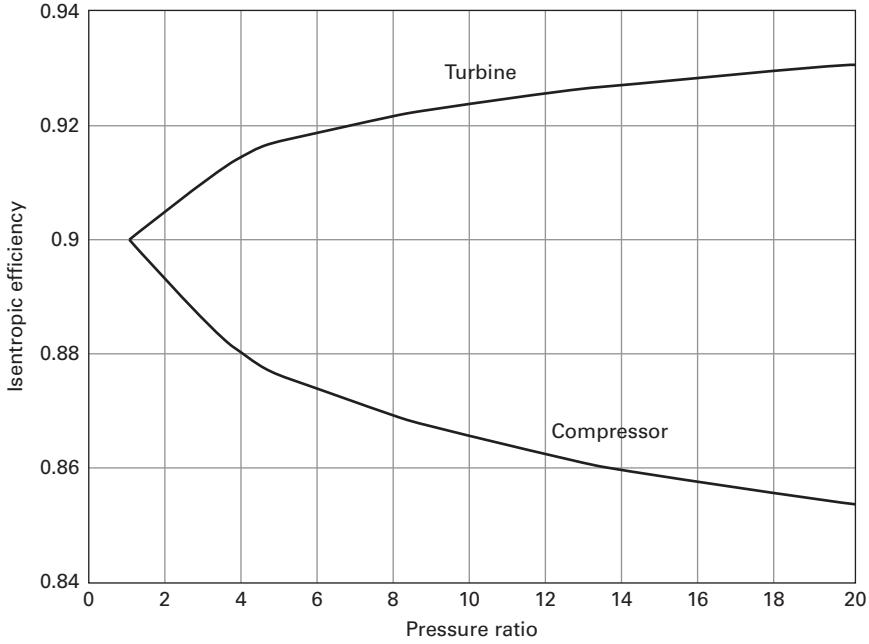


FIG. 2.12 Variation of turbine and compressor isentropic efficiency with pressure ratio for polytropic efficiency of 90 per cent

them in cycle calculations, the most convenient equations to use will be shown to be those corresponding to (2.11) and (2.12), i.e. from equations (2.15) and (2.17),

$$T_{02} - T_{01} = T_{01} \left[\left(\frac{p_{02}}{p_{01}} \right)^{(n-1)/n} - 1 \right] \quad (2.19)$$

where $(n-1)/n = (\gamma-1)/\gamma\eta_{\infty c}$, and

$$T_{03} - T_{04} = T_{03} \left[1 - \left(\frac{1}{p_{03}/p_{04}} \right)^{(n-1)/n} \right] \quad (2.20)$$

where $(n-1)/n = \eta_{\infty t}(\gamma-1)/\gamma$.

Again, for a compressor of an industrial gas turbine we shall take $p_{01} = p_a$ and $T_{01} = T_a$, while for turbines exhausting to atmosphere p_{04} will be put equal to p_a .

We end this sub-section with a reminder that the isentropic and polytropic efficiencies present the same information in different forms. When performing calculations over a range of pressure ratio, it is reasonable to assume constant polytropic efficiency; this automatically allows for a variation of isentropic efficiency with pressure ratio. In simple terms, the polytropic efficiency may be interpreted as representing the current state of the art for a particular design organization. When determining the performance of a single cycle of interest, or analysing engine test data, it is more appropriate to use isentropic efficiencies. Cycle

calculations based on isentropic efficiency should use equations (2.11) and (2.12) while those based on polytropic efficiency should use (2.19) and (2.20).

Pressure losses

Pressure losses in the intake and exhaust ducting have been dealt with in the previous sub-section. In the combustion chamber a loss in stagnation pressure (Δp_b) occurs owing to the aerodynamic resistance of flame-stabilizing and mixing devices, and also to momentum changes produced by the exothermic reaction. These sources of loss are referred to in detail in Chapter 6. When a heat-exchanger is included in the plant there will also be frictional pressure losses in the passages on the air side (Δp_{ha}) and gas side (Δp_{hg}). As shown in Fig. 2.13, the pressure losses have the effect of decreasing the turbine pressure ratio relative to the compressor pressure ratio and thus reduce the net work output from the plant. The gas turbine cycle is very sensitive to irreversibilities, because the net output is the difference of two large quantities (i.e. the 'work ratio' is low), so that the pressure losses have a significant effect on the cycle performance.

Fixed values of the losses can be fed into the cycle calculation directly. For example, for a simple cycle with heat-exchange we may determine the turbine pressure ratio p_{03}/p_{04} from

$$p_{03} = p_{02} - \Delta p_b - \Delta p_{ha} \quad \text{and} \quad p_{04} = p_a + \Delta p_{hg}$$

But again the question arises as to whether it is reasonable to assume constant values for the pressure drops when cycles of different pressure ratio are being compared. The frictional pressure losses will be roughly proportional to the local dynamic head of the flow ($\frac{1}{2}\rho C^2$ for incompressible flow), as with ordinary pipe flow. It might therefore be expected that the pressure drops Δp_{ha} and Δp_b will increase with cycle pressure ratio because the density of the fluid in the air side

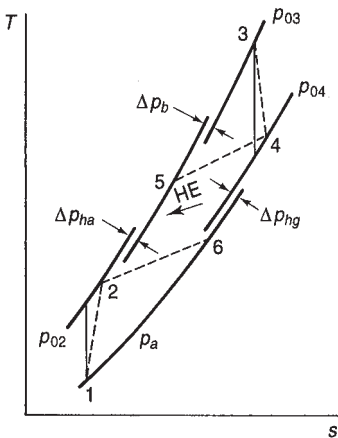


FIG. 2.13 Pressure losses

of the heat-exchanger and in the combustion chamber is increased. Even though ρ is not proportional to p because T increases also, a better approximation might be to take Δp_{ha} and Δp_b as fixed proportions of the compressor delivery pressure. The turbine inlet pressure is then found from

$$p_{03} = p_{02} \left(1 - \frac{\Delta p_b}{p_{02}} - \frac{\Delta p_{ha}}{p_{02}} \right)$$

We shall express the pressure loss data in this way in subsequent numerical examples.

The pressure loss in the combustor can be minimized by using a large combustion chamber with consequent low velocities, which is feasible in an industrial unit where size is not critical. For an aero engine, where weight, volume and frontal area are all important, higher pressure losses are inevitable. The designer also has to consider pressure losses at a typical flight condition at high altitude, where the air density is low. Typical values of $\Delta p_b/p_{02}$ would be about 2–3 per cent for a large industrial unit and 6–8 per cent for an aero engine.

Heat-exchanger effectiveness

Although the thermodynamic merits of heat-exchangers have been recognized from the earliest days of gas turbines, it should be emphasized that the vast majority of industrial gas turbines have relied on continuous increase in pressure ratio to achieve high efficiency.

Heat-exchangers for gas turbines can take many forms, including counterflow and cross-flow recuperators (where the hot and cold streams exchange heat through a separating wall) or regenerators (where the streams are brought cyclically into contact with a matrix which alternately absorbs and rejects heat). In all cases, using the notation of Fig. 2.13, the fundamental process is that the turbine exhaust gases reject heat at the rate of $m_t c_{p46}(T_{04} - T_{06})$ while the compressor delivery air receives heat at the rate of $m_c c_{p25}(T_{05} - T_{02})$. For conservation of energy, assuming that the mass flows m_t and m_c are equal,

$$c_{p46}(T_{04} - T_{06}) = c_{p25}(T_{05} - T_{02}) \quad (2.21)$$

But both T_{05} and T_{06} are unknown and a second equation is required for their evaluation. This is provided by the equation expressing the efficiency of the heat-exchanger.

Now the maximum possible value of T_{05} is when the ‘cold’ air attains the temperature of the incoming hot gas T_{04} , and one possible measure of performance is the ratio of actual energy received by the cold air to the maximum possible value, i.e.

$$\frac{m_c c_{p25}(T_{05} - T_{02})}{m_c c_{p24}(T_{04} - T_{02})}$$

The mean specific heat of air will not be very different over the two temperature ranges, and it is usual to define the efficiency in terms of temperature alone and to call it the *effectiveness* (or *thermal ratio*) of the heat-exchanger. Thus

$$\text{effectiveness} = \frac{T_{05} - T_{02}}{T_{04} - T_{02}} \quad (2.22)$$

When a value of effectiveness is specified, equation (2.22) enables the temperature at the inlet to the combustion chamber T_{05} to be determined.

Equation (2.21) then yields T_{06} if required. Note that the mean specific heats c_{p46} and c_{p25} are not approximately equal and cannot be cancelled, because the former is for turbine exhaust gas and the latter for air.

In general, the larger the volume of the heat-exchanger, the higher the effectiveness, but the increase in effectiveness becomes asymptotic beyond a certain value of heat transfer surface area; the cost of heat-exchangers is largely determined by their surface area, and with this in mind modern heat-exchangers are designed to have an effectiveness of about 0.90. The maximum permissible turbine *exit* temperature is fixed by the materials used for the construction of the heat-exchanger. With stainless steel, this temperature should not be much above 900 K. It follows that the turbine inlet temperature of a regenerative cycle is limited for this reason. Heat-exchangers are subjected to severe thermal stresses during start-up and are not used where frequent starts are required, e.g. for peak-load electricity generation. They are best suited for applications where the gas turbine operates for extended periods at a steady power, as may occur on a pipeline. It should be noted, however, that many pipeline stations are located in remote areas where transportation and installation of a heat-exchanger present major problems; for this reason, some modern heat-exchangers are built up from a series of identical modules. In recent years heat-exchangers have seldom been used, because they no longer offer an advantage relative to simple cycles of high pressure ratio or, in the case of base-load applications, relative to combined cycle plant. In the late 1990s, however, a high-efficiency small electric power unit of 4 MW, the Solar Mercury, was introduced. This engine was designed from the outset to be built with a heat-exchanger, and the design of the turbomachinery and combustor was configured to provide the most convenient flow paths to and from the heat-exchanger. The resulting layout is shown in Fig. 2.14, where it can be seen that the inlet to the compressor is located near the centre of the engine and the combustor is located at one end; this enables combustor modifications to be easily incorporated, e.g. for use with different fuels or as new low-emission technologies are developed. If microturbines are successfully introduced to the market, they will undoubtedly require heat-exchangers to give a reasonable cycle efficiency because of their low pressure ratios; they would be expected to be used in cogeneration systems where an HRSG would produce steam or hot water for process use. A microturbine self-contained package is shown in Fig. 2.15, which clearly shows the very small size of the actual turbomachinery and the relatively large size of the heat-exchanger. The future search for cycle efficiencies exceeding 60 per cent could see the return of the heat-exchanger in complex cycles, and a survey of the possibilities is given in Ref. (2).

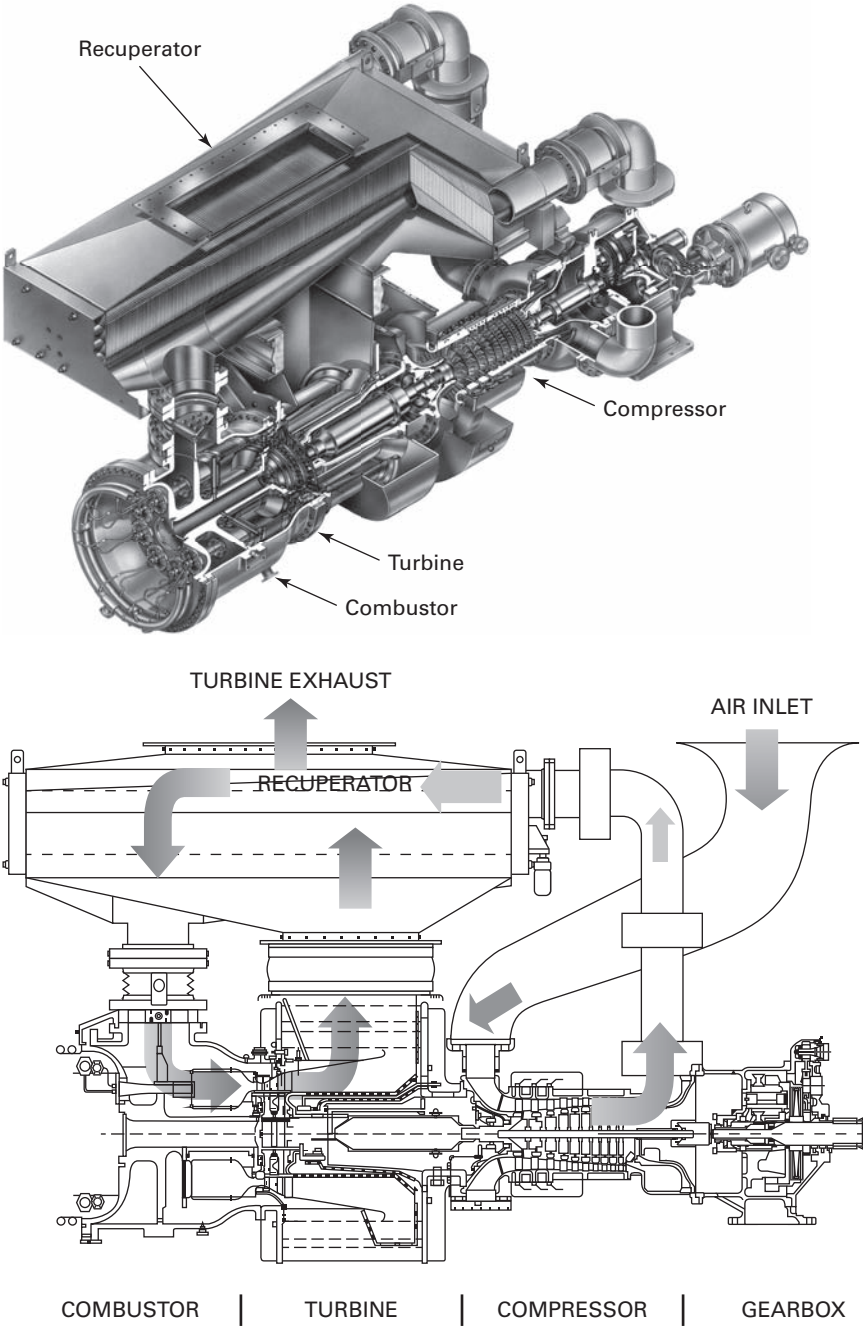


FIG. 2.14 Solar Mercury gas turbine [courtesy Solar Turbines International]

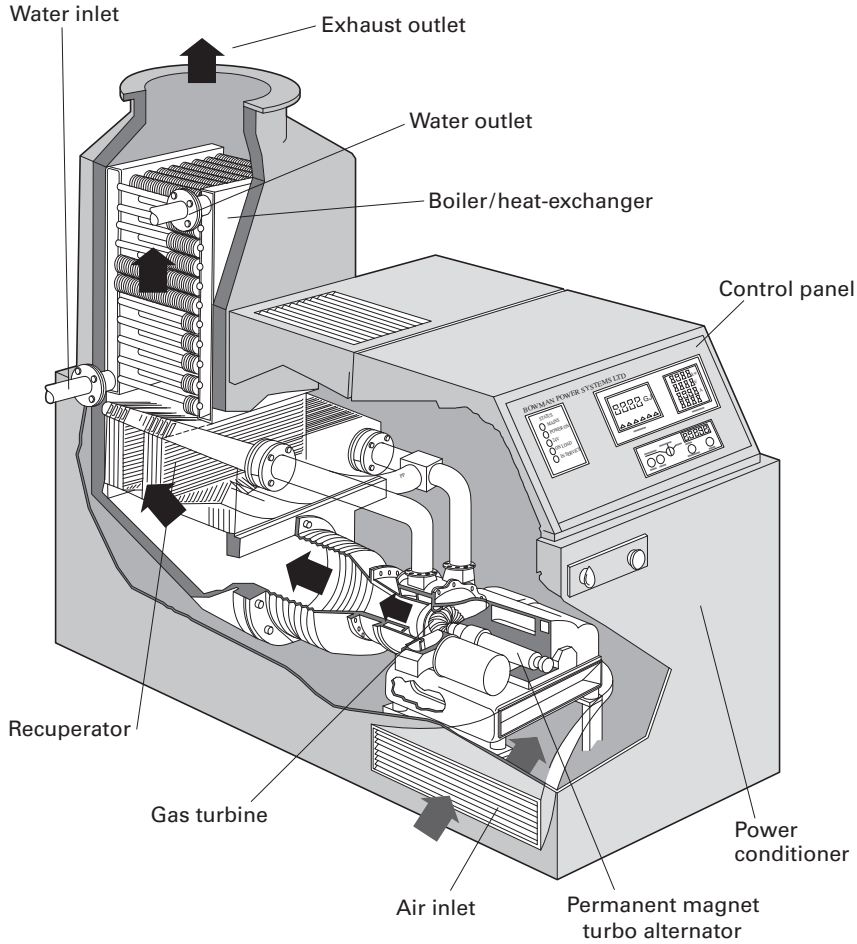


FIG. 2.15 Microturbine package [courtesy Bowman Power Systems]

Mechanical losses

In all gas turbines, the power necessary to drive the compressor is transmitted directly from the turbine without any intermediate gearing.

Any loss that occurs is therefore due only to bearing friction and windage. This loss is very small and it is normal to assume that it amounts to about 1 per cent of the power necessary to drive the compressor. If the transmission efficiency is denoted by η_m , we have the work output required to drive the compressor given by

$$W = \frac{1}{\eta_m} c_{p12} (T_{02} - T_{01})$$

We shall take η_m to be 99 per cent for all numerical examples.

Any power used to drive ancillary components such as fuel and oil pumps can often be accounted for simply by subtracting it from the net output of the unit. Power absorbed in any gearing between the gas turbine and the load can be dealt with similarly. Except to say that such losses can be significant, especially for small gas turbines of low power, we shall not consider them further. Their consideration would not require the method of cycle performance estimation to be modified except when the gas turbine has a separate power turbine. Power for fuel and oil pumps will then be taken from the compressor turbine (because under some operating conditions the power turbine is stationary).

Variation of specific heat

The properties c_p and γ play an important part in the estimation of cycle performance, and it is necessary to take account of variations in values due to changing conditions through the cycle. In general, for real gases over normal working ranges of pressure and temperature, c_p is a function of temperature alone. The same is true of γ because it is related to c_p by

$$\frac{\gamma - 1}{\gamma} = \frac{\tilde{R}}{Mc_p} \quad (2.23)$$

where $\tilde{R}$ is the molar (universal) gas constant and M the molecular weight. The variation of c_p and γ with temperature for air is shown in Fig. 2.16 by the curves marked zero fuel/air ratio. Only the left-hand portion is of interest because even with a pressure ratio as high as 35 the compressor delivery temperature will be no more than about 800 K.

In the turbine of an open-cycle plant the working fluid will be a mixture of combustion gases. Most gas turbines run on kerosene, which has a composition to which the formula C_nH_{2n} is a good approximation. If some such composition is assumed, the products analysis can be calculated for various fuel/air ratios. Knowing the specific heats and molecular weights of the constituents it is then a simple matter to calculate the mean values of c_p and γ for the mixture. Figure 2.16 shows that c_p increases and γ decreases with increase in fuel/air ratio. It is worth noting that the mean molecular weight of the combustion products from typical hydrocarbon fuels is little different from that of air, and therefore c_p and γ are closely related by equation (2.23) with $\tilde{R}/M = R_{\text{air}} = 0.287 \text{ kJ/kg K}$.

The calculation of the product analyses is very lengthy when dissociation is taken into account and then, because pressure has a significant effect on the amount of dissociation, c_p and γ become a function of pressure as well as temperature. Accurate calculations of this kind have been made, and the results are tabulated in Ref. (3). Dissociation begins to have a significant effect on c_p and γ at a temperature of about 1500 K, and above this temperature the curves of Fig. 2.16 are strictly speaking applicable only to a pressure of 1 bar. In fact at 1800 K, both for air and products of combustion corresponding to low values of fuel/air ratio, a reduction of pressure to

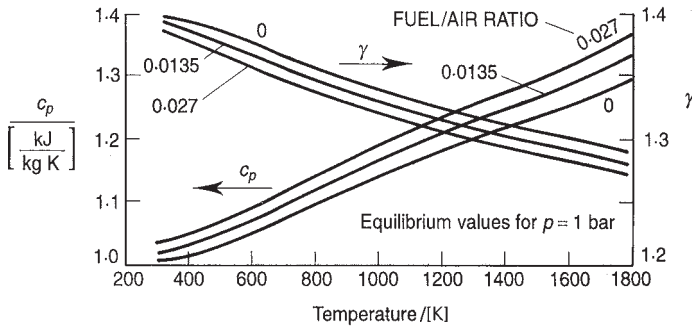


FIG. 2.16 c_p and γ for air and typical combustion gases

0.01 bar increases c_p by only about 4 per cent and an increase to 100 bar decreases c_p by only about 1 per cent: the corresponding changes in γ are even smaller. In this book we will ignore any effect of pressure, although many aircraft and industrial gas turbines are designed to use turbine inlet temperatures in excess of 1500 K.

Now compressor temperature rises and turbine temperature drops will be calculated using equations such as (2.11) and (2.12), or (2.19) and (2.20). For accurate calculations a method of successive approximation would be required, i.e. it would be necessary to guess a value of γ , calculate the temperature change, take a more accurate mean value of γ and recalculate the temperature change. In fact, if this degree of accuracy is required it is better to use tables or curves of enthalpy and entropy as described, for example, in Ref. (1). For preliminary design calculations and comparative cycle calculations, however, it has been found to be sufficiently accurate to assume the following fixed values of c_p and γ for the compression and expansion processes respectively:

$$\begin{aligned} \text{air:} \quad c_{pa} &= 1.005 \text{ kJ/kg K, } \gamma_a = 1.40 \text{ or } \left(\frac{\gamma}{\gamma - 1} \right)_a = 3.5 \\ \text{combustion gases:} \quad c_{pg} &= 1.148 \text{ kJ/kg K, } \gamma_g = 1.333 \text{ or } \left(\frac{\gamma}{\gamma - 1} \right)_g = 4.0 \end{aligned}$$

The reason why this does not lead to much inaccuracy is that c_p and γ vary in opposing senses with T . For cycle analysis we are interested in calculating compressor and turbine work from the product $c_p \Delta T$. Suppose that the temperature for which the above values of c_p and γ are the true values is lower than the actual mean temperature. γ is then higher than it should be and ΔT will be overestimated. This will be compensated in the product $c_p \Delta T$ by the fact that c_p will be lower than it should be. The actual temperatures at various points in the cycle will not be very accurate, however, and for the detailed design of the components it is necessary to know the exact conditions of the working fluid: the more accurate iterative approaches mentioned above must then be employed.

Fuel/air ratio, combustion efficiency and cycle efficiency

The performance of real cycles can be unambiguously expressed in terms of the specific fuel consumption, i.e. fuel mass flow per unit net power output. To obtain this the fuel/air ratio must be found. In the course of calculating the net output per unit mass flow of air, the temperature at the inlet to the combustion chamber (T_{02}) will have been obtained; and the temperature at the outlet (T_{03}), which is the maximum cycle temperature, will normally be specified. The problem is therefore to calculate the fuel/air ratio f required to transform unit mass of air at T_{02} and f kg of fuel at the fuel temperature T_f to $(1 + f)$ kg of products at T_{03} .

Since the process is adiabatic with no work transfer, the energy equation is simply

$$\sum (m_i h_{i03}) - (h_{a02} + f h_f) = 0$$

where m_i is the mass of product i per unit mass of air and h_i its specific enthalpy. Making use of the enthalpy of reaction at a reference temperature of 25°C, ΔH_{25} , the equation can be expanded in the usual way [see Ref. (1)] to become

$$(1 + f)c_{pg}(T_{03} - 298) + f\Delta H_{25} + c_{pa}(298 - T_{02}) + fc_{pf}(298 - T_f) = 0$$

where c_{pg} is the mean specific heat of the products over the temperature range 298 K to T_{03} . ΔH_{25} should be the enthalpy of reaction per unit mass of fuel with the H_2O in the products in the vapour phase, because T_{03} is high and above the dew point. For common fuels ΔH_{25} may be taken from tables, or alternatively it may be evaluated from the enthalpies of formation of the reactants. It is usual to assume that the fuel temperature is the same as the reference temperature, so that the fourth term on the LHS of the equation is zero. The term will certainly be small because f is low (≈ 0.02) and c_{pf} for liquid hydrocarbon fuels is only about 2 kJ/kgK. Finally, we have already discussed in the previous sub-section, 'Variation of specific heat', the calculation of the mean specific heat of the products c_{pg} as a function of f and T , so we are left with an equation from which f can be obtained for any given values of T_{02} and T_{03} .

Such calculations are too lengthy to be undertaken for every individual cycle calculation—particularly if dissociation is significant, because then the $f\Delta H_{25}$ term must be modified to allow for the incompletely burnt carbon and hydrogen arising from the dissociated CO_2 and H_2O . It is usually sufficiently accurate to use tables or charts which have been compiled for a typical fuel composition. Figure 2.17 shows the combustion temperature rise ($T_{03} - T_{02}$) plotted versus fuel/air ratio for various values of inlet temperature (T_{02}), and these curves will be used for all numerical examples in this book. It is a small-scale version of larger and more accurate graphs given in Ref. (4). The reference fuel for which the data have been calculated is a hypothetical liquid hydrocarbon containing 13.92 per cent H and 86.08 per cent C, for which the stoichiometric fuel/air ratio is 0.068 and ΔH_{25} is $-43\,100$ kJ/kg. The curves are certainly adequate for any kerosene burnt in dry air. Methods are given in Ref. (4) whereby the data can be used for hydrocarbon fuels which differ widely in composition from the reference fuel, or where the fuel

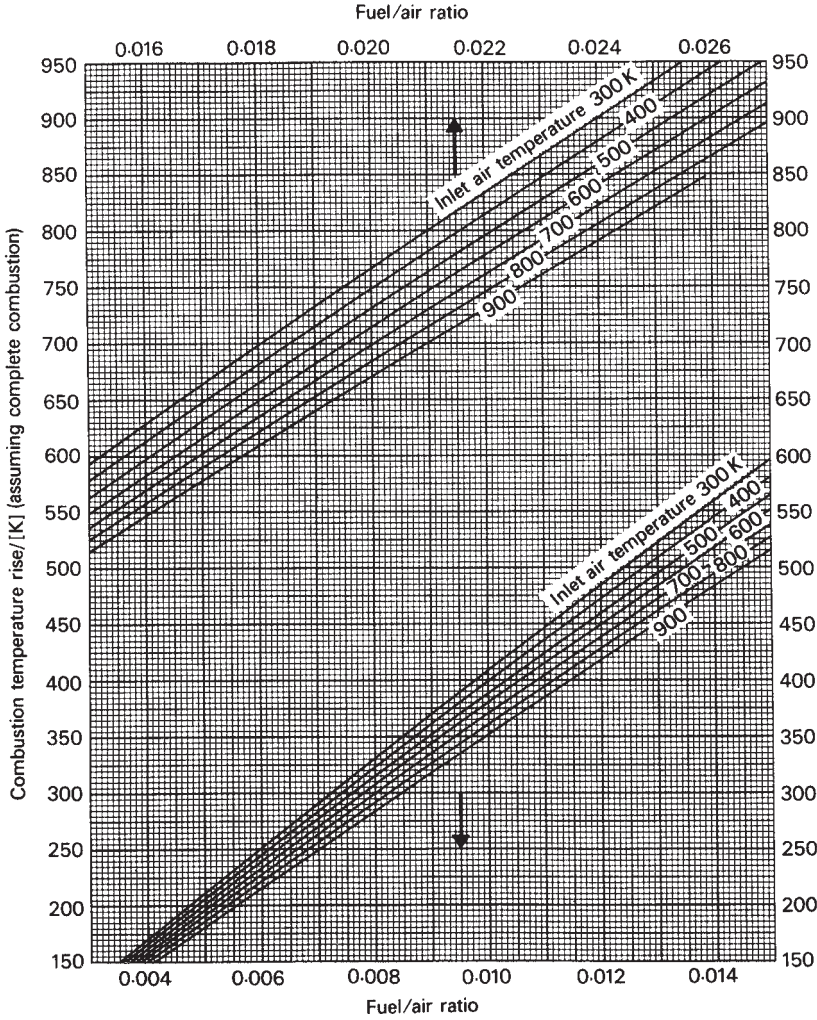


FIG. 2.17 Combustion temperature rise versus fuel/air ratio

is burnt in a reheat chamber, i.e. not in air but in the products of combustion from a previous chamber in the cycle. Chapter 6.2 revisits the calculation of fuel/air ratio, and provides more detail on chemical balancing of combustion reactions and calculation of flame temperature.

The data for Fig. 2.17 have been calculated on the assumption that the fuel is completely burnt, and thus the abscissa could be labelled 'theoretical fuel/air ratio'. The most convenient method of allowing for combustion loss is by introducing a *combustion efficiency* defined by

$$\eta_b = \frac{\text{theoretical } f \text{ for given } \Delta T}{\text{actual } f \text{ for given } \Delta T}$$

This is the definition used in this book. An alternative method is to regard the ordinate as the theoretical ΔT for a given f and define the efficiency in terms of the ratio: actual ΔT /theoretical ΔT . Neither definition is quite the same as the fundamental definition based on the ratio of actual energy released to the theoretical quantity obtainable. (η_b differs from it because of the small additional heat capacity of the products arising from the increase in fuel needed to produce the given temperature.) But in practice combustion is so nearly complete—98–99 per cent—that the efficiency is difficult to measure accurately and the three definitions of efficiency yield virtually the same result.

Once the fuel/air ratio is known, the fuel consumption m_f is simply $f \times m$ where m is the air mass flow, and the specific fuel consumption can be found directly from

$$SFC = \frac{f}{W_N}$$

Since the fuel consumption is normally measured in kg/h, while W_N is in kW per kg/s of airflow, the SFC in kg/kWh is given by the following numerical equation:†

$$\frac{SFC}{[\text{kg/kW h}]} = \frac{f}{W_N/[\text{kW s/kg}]} \times \frac{[\text{s}]}{[\text{h}]} = \frac{3600 f}{W_N/[\text{kW s/kg}]}$$

When dealing with a specific engine rather than cycle calculations it is more convenient to calculate the SFC from

$$\frac{SFC}{[\text{kg/kW h}]} = \frac{m_f}{\text{power}}$$

If the thermal efficiency of the cycle is required, it must be defined in the form ‘work output/heat supplied’ even though the combustion process is adiabatic and in the thermodynamic sense no heat is supplied. We know that if the fuel is burnt under ideal conditions, such that the products and reactants are virtually at the same temperature (the reference temperature 25°C), the rate of energy release in the form of heat will be

$$m_f Q_{\text{gr},p} = f m Q_{\text{gr},p}$$

where m_f is the fuel flow and $Q_{\text{gr},p}$ is the gross (or higher) calorific value at constant pressure. In the gas turbine it is not possible to utilize the latent heat of the H_2O vapour in the products and the convention of using the net calorific value has been adopted in most countries. Thus the cycle efficiency may be defined as

$$\eta = \frac{W_N}{f Q_{\text{net},p}}$$

†The fundamental principle employed here is that a physical quantity, or the symbol representing it, is equal to (pure number $\times$ unit) and *never* the number alone. Consequently an equation relating numbers only will, in addition to pure numbers and dimensional ratios, contain only quotients of symbols and units, e.g. $W_N/[\text{kW s/kg}]$.

With the units used here, the equivalent numerical equation becomes

$$\eta = \frac{W_N / [\text{kW s/kg}]}{f \times Q_{\text{net},p} / [\text{kJ or kW s/kg}]}$$

or, using the above numerical equation for SFC ,

$$\eta = \frac{3600}{SFC / [\text{kg/kW h}] \times Q_{\text{net},p} / [\text{kJ/kg}]}$$

$Q_{\text{net},p}$ is sensibly equal in magnitude but opposite in sign to the enthalpy of reaction ΔH_{25} referred to earlier, and the value of 43 100 kJ/kg will be used for all numerical examples.

When referring to the thermal efficiency of actual gas turbines, manufacturers often prefer to use the concept of *heat rate* rather than efficiency. The reason is that fuel prices are normally quoted in terms of pounds sterling (or dollars) per megajoule, and the heat rate can be used to evaluate fuel cost directly. Heat rate is defined as $(SFC \times Q_{\text{net},p})$, and thus expresses the heat input required to produce a unit quantity of power. It is normally expressed in kJ/kW h, in which case the corresponding thermal efficiency can be found from $3600/(\text{heat rate})$.

Bleed flows

In section 2.1 the importance of high turbine inlet temperature was demonstrated. Turbine inlet temperatures, however, are limited by metallurgical considerations and many modern engines make use of air-cooled blades to permit operation at elevated temperatures. It is possible to operate with uncooled blades up to about 1350–1400 K; and then it can be assumed that the mass flow remains constant throughout as explained earlier. At higher temperatures it is necessary to extract air to cool both stator and rotor blades. The required bleeds may amount to 15 per cent or more of the compressor delivery flow in an advanced engine, and must be properly accounted for in accurate calculations. The overall air cooling system for such an engine is complex, but the methods of dealing with bleeds can be illustrated using a simplified example. A single-stage turbine with cooling of both the stator and rotor blades will be considered, as shown schematically in Fig. 2.18; β_D , β_S and β_R denote bleeds for cooling the disc, stator and rotor respectively. Bleeds are normally specified as a percentage of the compressor delivery flow.

Note that bleed β_D acts to prevent the flow of hot gases down the face of the turbine disc, but it does pass through the rotor. The stator bleed also passes through the rotor and both contribute to the power developed. The rotor bleed, however, does not contribute to the work output and the reduction in mass flow results in an increase in both temperature drop and pressure ratio for the specified power. Useful work would be done by the rotor bleed only if there were another turbine downstream.

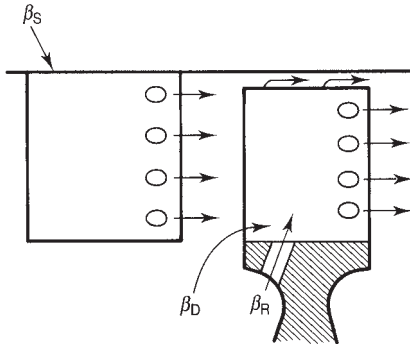


FIG. 2.18 Cooling air schematic

If the airflow at compressor delivery is m_a , then the flow available to the rotor, m_R , is given by

$$m_R = m_a(1 - \beta_R) + m_f$$

The fuel flow is found from the fuel/air ratio required for the given combustion temperature rise and combustion inlet temperature, and the air available for combustion, i.e.

$$m_a(1 - \beta_D - \beta_S - \beta_R)$$

Hence the fuel flow is given by

$$m_f = m_a(1 - \beta_D - \beta_S - \beta_R) \cdot f$$

It should be noted that the cooling flows from the stator and disc, at the compressor delivery temperature, will cause some reduction in the effective temperature at entry to the rotor. This effect can be estimated by carrying out an enthalpy balance assuming complete mixing of the flows. Such perfect mixing does not occur in practice, but accurate cycle calculations would require a good estimate to be made of this cooling of the main flow.

In a typical cooled stator with a β_S of 6 per cent, the stator outlet temperature may be reduced by about 100 K. Since the rotor inlet temperature is equal to the stator outlet temperature, this implies a reduction in turbine power output. In addition there will be a small drop in efficiency due to the mixing of the bleeds with the main stream.

The distribution of cooling air in a hypothetical multi-stage turbine is shown in Fig. 2.19, which gives some idea of the complexity and the need to understand fully the location of the various bleeds.

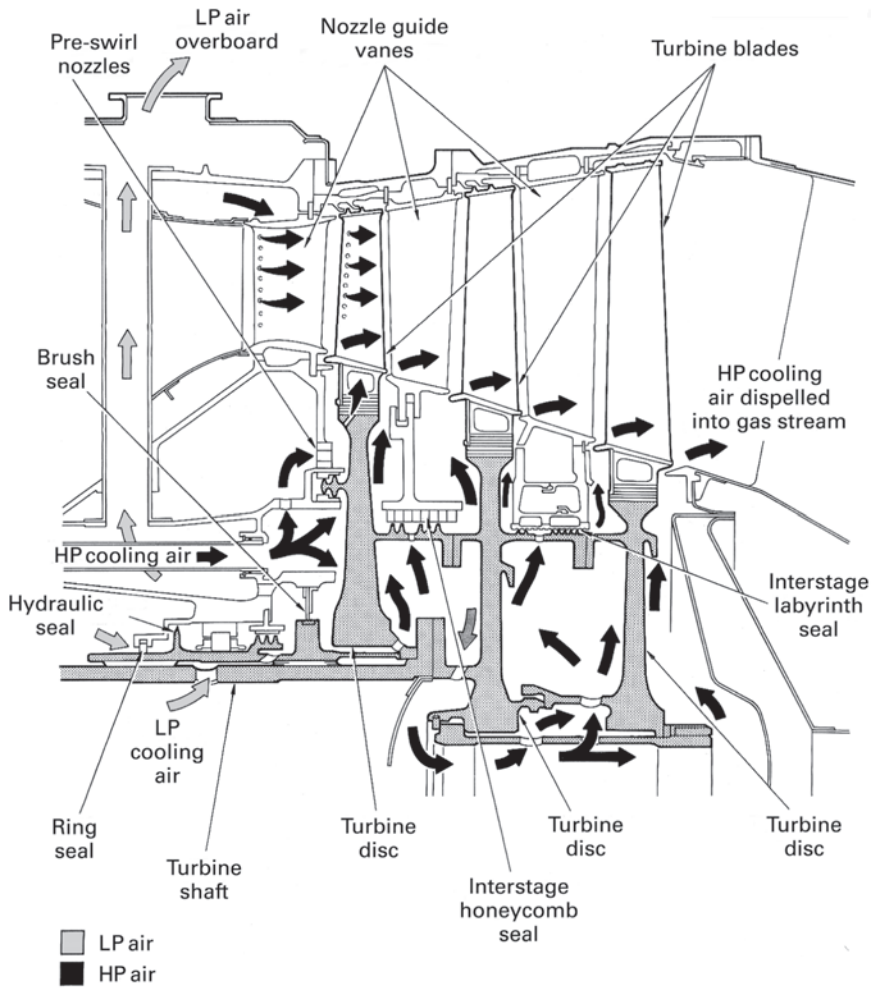


FIG. 2.19 Cooling air distribution [courtesy Rolls-Royce plc]

2.3 Design point performance calculations

Before assessing the effect of component losses on the general performance curves for the various cycles considered in section 2.1, it is necessary to outline the method of calculating the performance in any particular case for specified values of the design parameters. These parameters will include the compressor pressure ratio, turbine inlet temperature (*TIT*), component efficiencies and pressure losses.